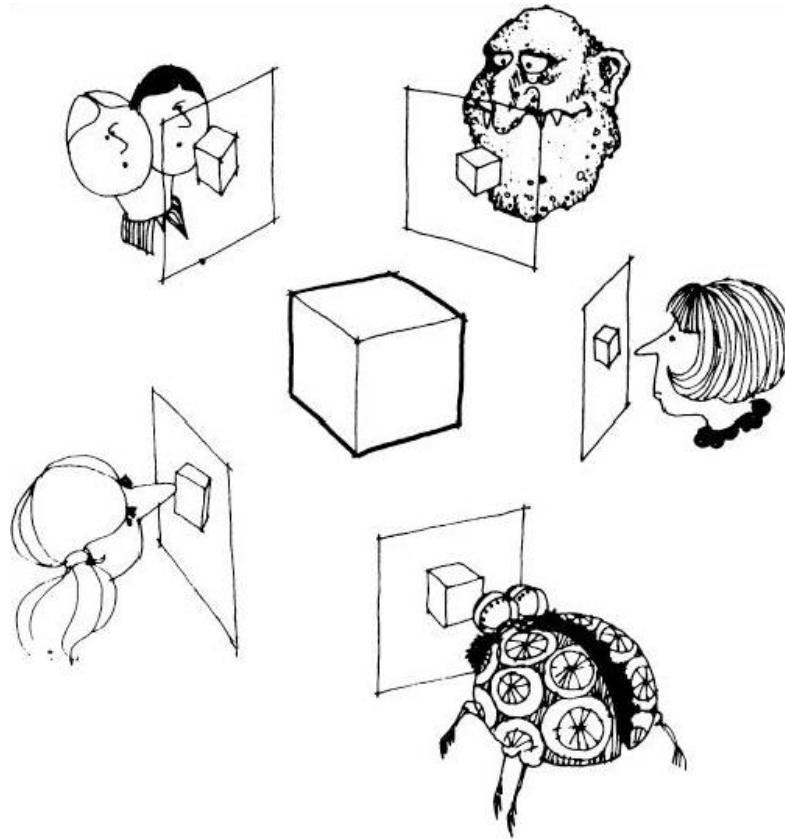


Perspective and 3D Geometry



Computer Vision

Adduru U G Sankararao, IIIT Sri City

Perspective and 3D Geometry

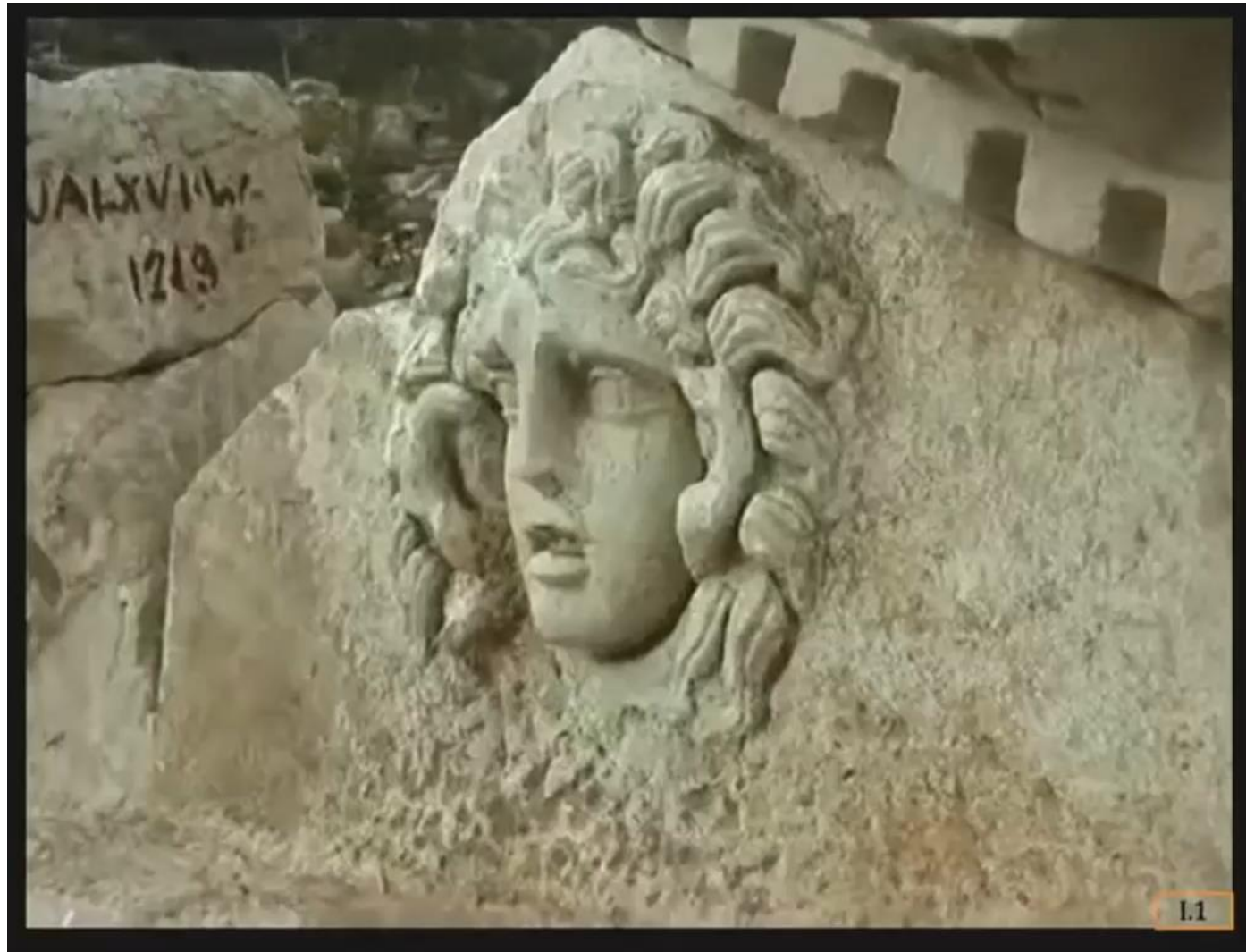
- **Structure from motion**

- How can we **recover 3D points from multiple images?**

Structure from Motion

Uncontrolled (Casual) Video

Method to estimate 3D structure of a static scene from two arbitrary views.



[Pollefeys 2002]

Structure From Motion

Compute 3D scene structure and camera motion from a sequence of frames.

- Structure from Motion Problem
- SFM Observation Matrix
- Rank of Observation Matrix
- Tomasi-Kanade Factorization.

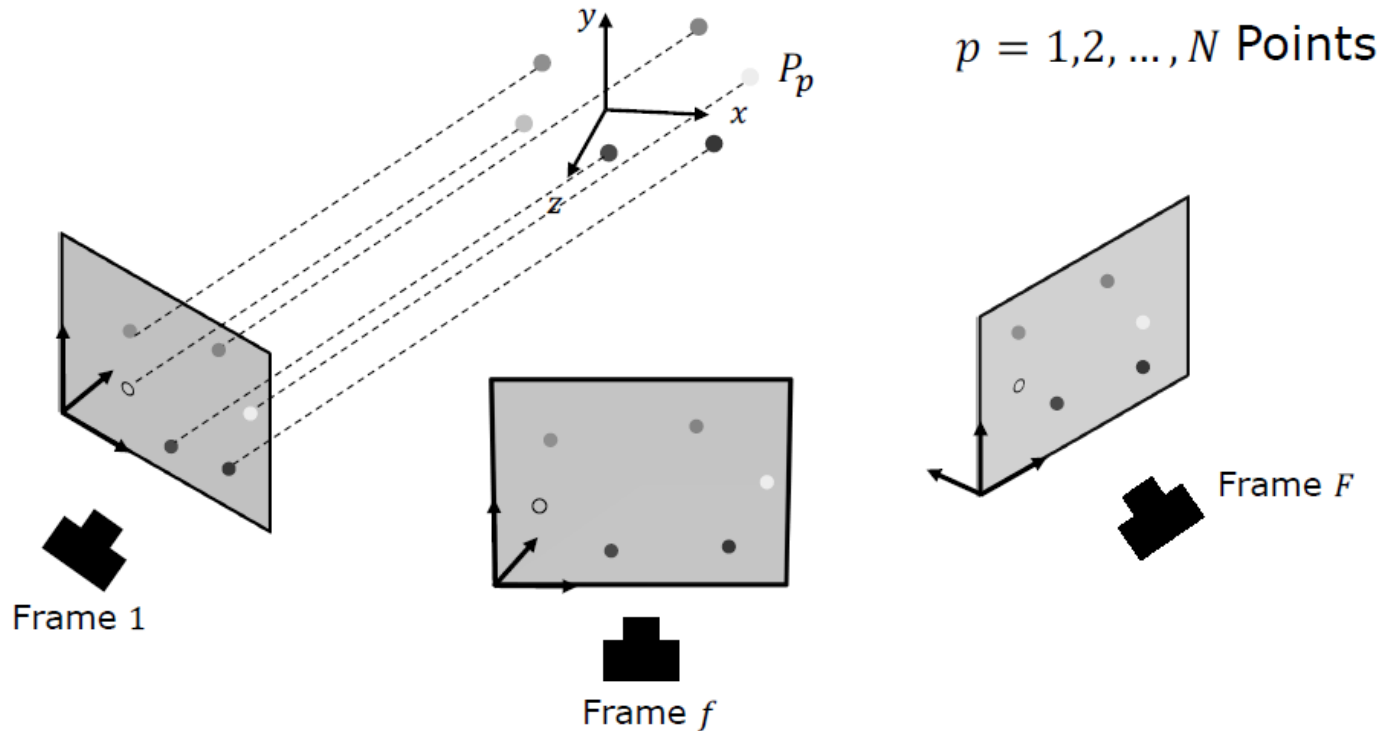
Structure from Motion: Problem

Feature Detection and Tracking

- Detect feature points: Corners, SIFT points, ...
- Track feature points: Template Matching, Optical Flow...
- The input to the SFM model is these feature coordinates



Orthographic Structure from Motion



Given sets of corresponding image points (2D): $(u_{f,p}, v_{f,p})$

Find scene points (3D) P_p , assuming orthographic camera.

Initial
Assumption

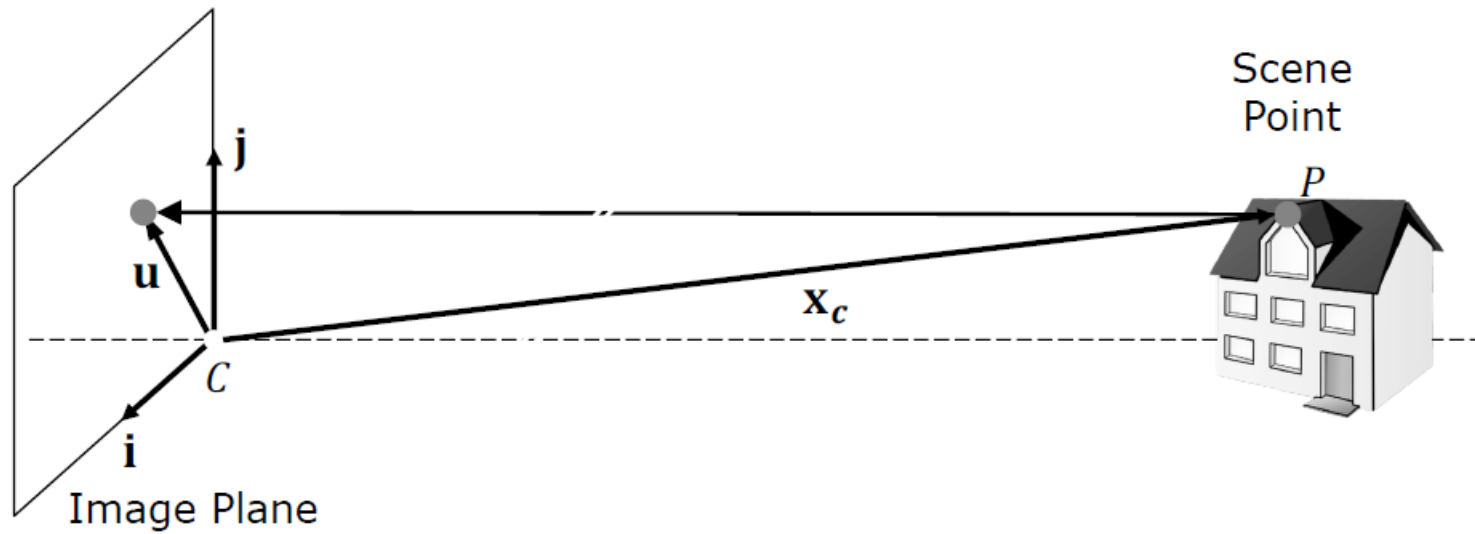
- To make this problem tractable, Tomasi and Kanade used the simplifying assumption that the camera is orthographic
- Orthographic: assume that the range of depths in the scene is small in comparison to the distance of the scene from the camera

[Tomasi 1992]

Observation Matrix

Organize all tracked image coordinates, $(u_{f,p}, v_{f,p})$ into a single matrix called the observation matrix.

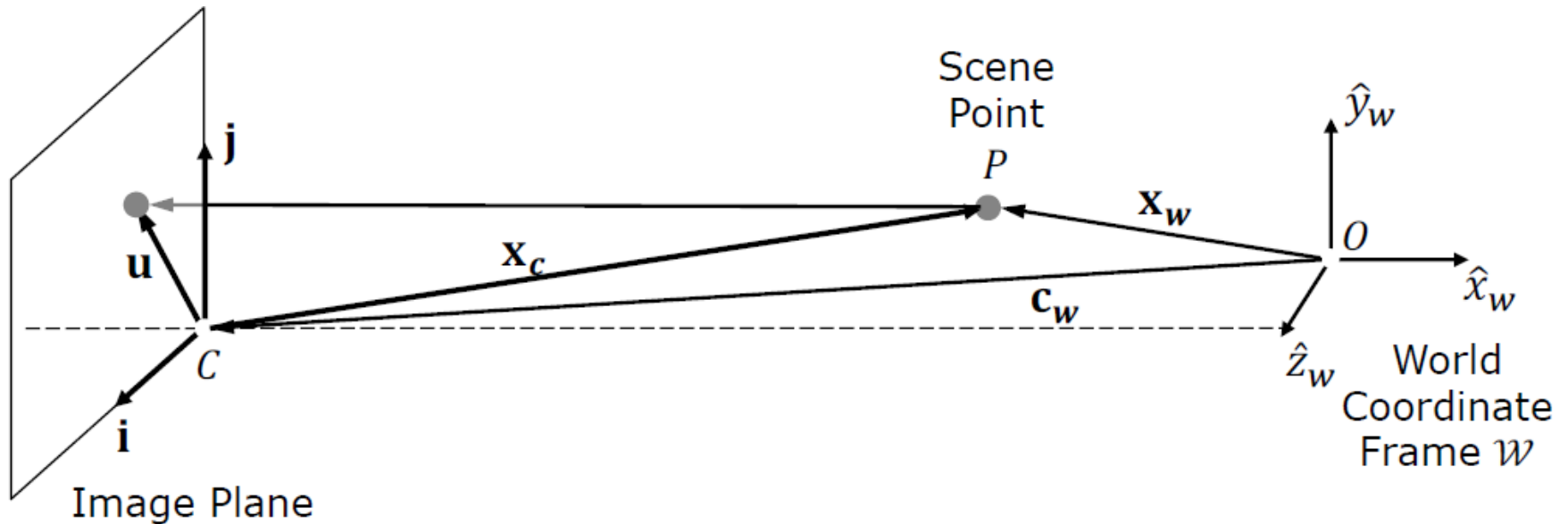
From 3D to 2D: Orthographic Projection



$$\begin{aligned} u &= \mathbf{i} \cdot \mathbf{x}_c = \mathbf{i}^T \mathbf{x}_c \\ v &= \mathbf{j} \cdot \mathbf{x}_c = \mathbf{j}^T \mathbf{x}_c \end{aligned} \quad \boxed{1}$$

Perspective cameras exhibit orthographic projection when distance of scene from camera is large compared to depth variation within scene (magnification is nearly constant).

From 3D to 2D: Orthographic Projection

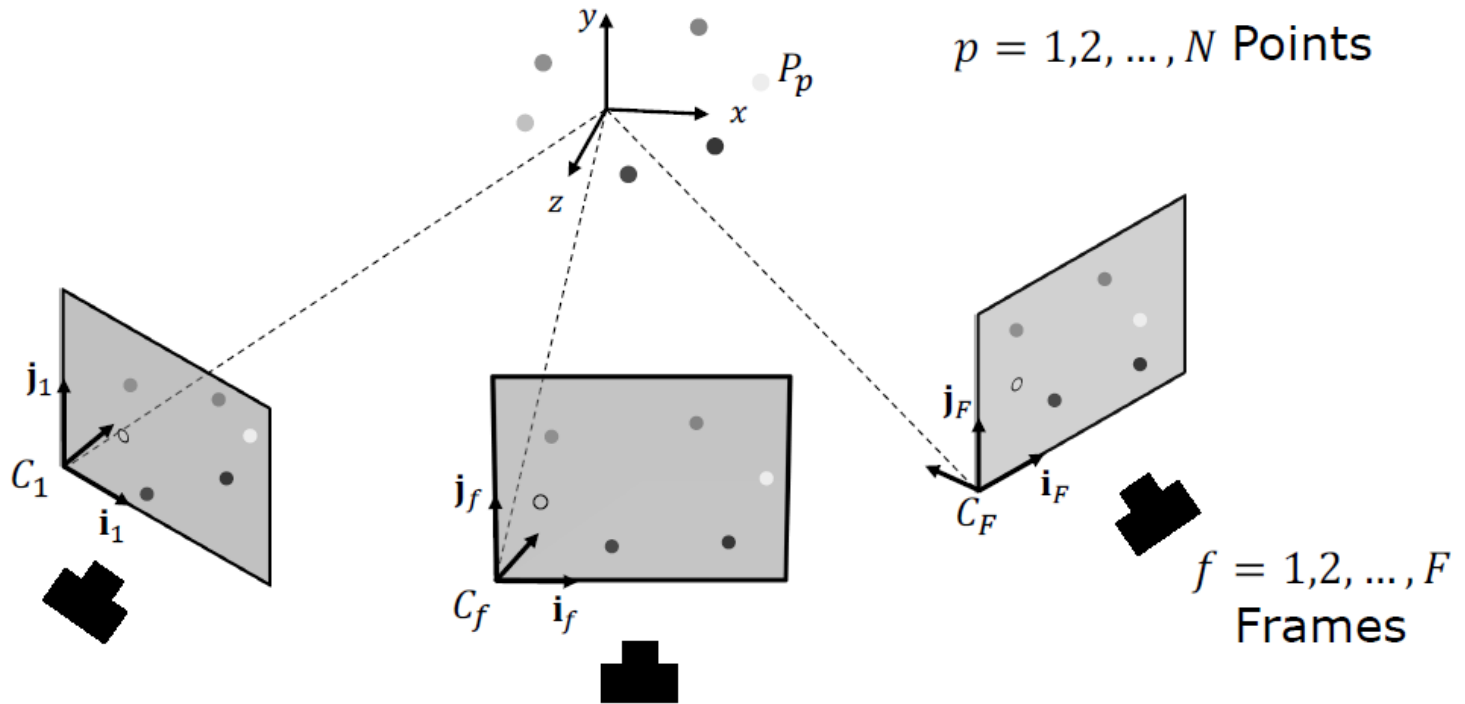


$$\begin{aligned} u &= \mathbf{i}^T \mathbf{x}_c = \mathbf{i}^T (\mathbf{x}_w - \mathbf{c}_w) = \mathbf{i}^T (P - C) \\ v &= \mathbf{j}^T \mathbf{x}_c = \mathbf{j}^T (\mathbf{x}_w - \mathbf{c}_w) = \mathbf{j}^T (P - C) \end{aligned} \quad \boxed{1}$$

$$\begin{aligned} u &= \mathbf{i}^T (P - C) \\ v &= \mathbf{j}^T (P - C) \end{aligned}$$

Gives the orthographic projection of any point in the scene to the camera.

Orthographic SFM



Given corresponding image points (2D) $(u_{f,p}, v_{f,p})$

Find scene points $\{P_p\}$.

Camera Positions $\{C_f\}$, camera orientations $\{(\mathbf{i}_f, \mathbf{j}_f)\}$ are unknown.

This is the Orthographic SFM problem

Orthographic SFM

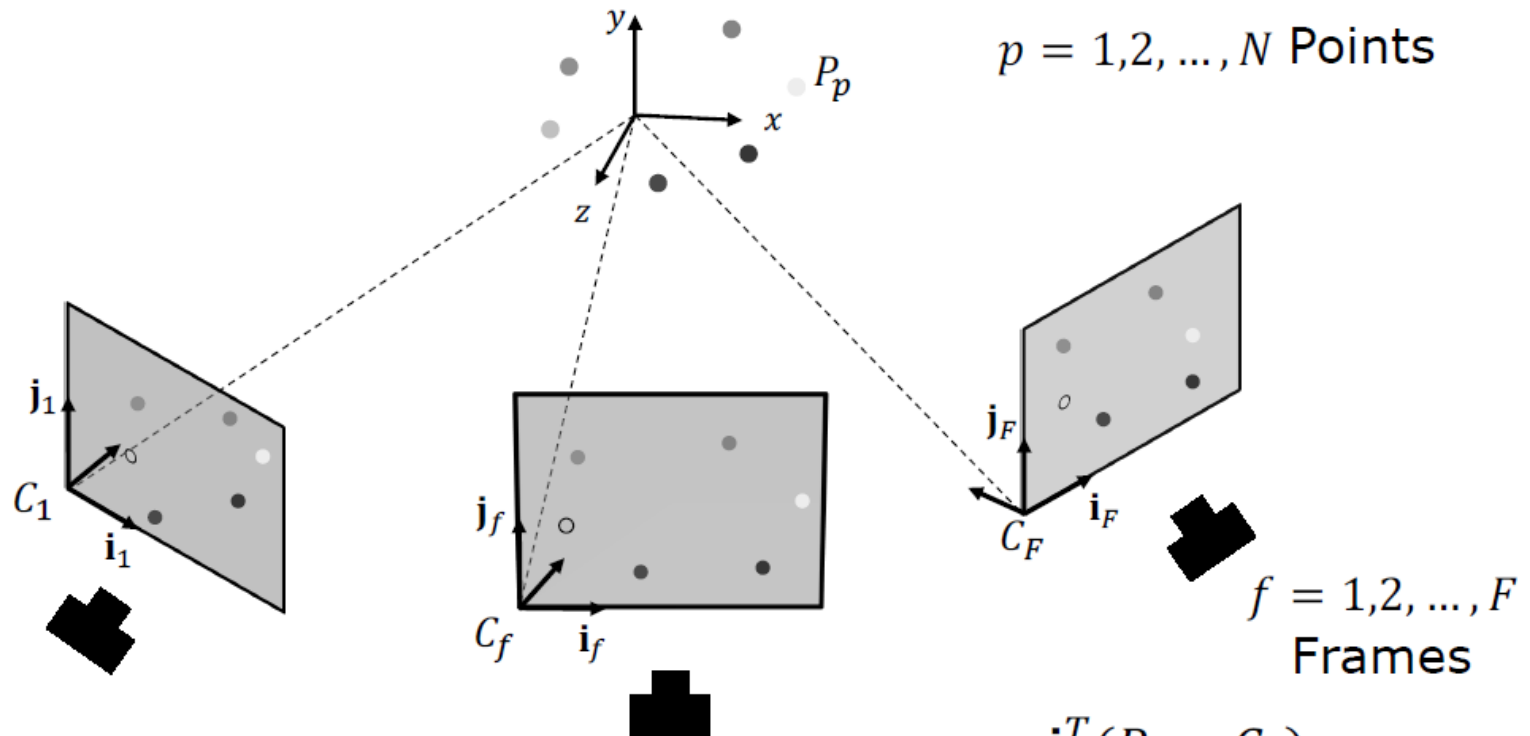


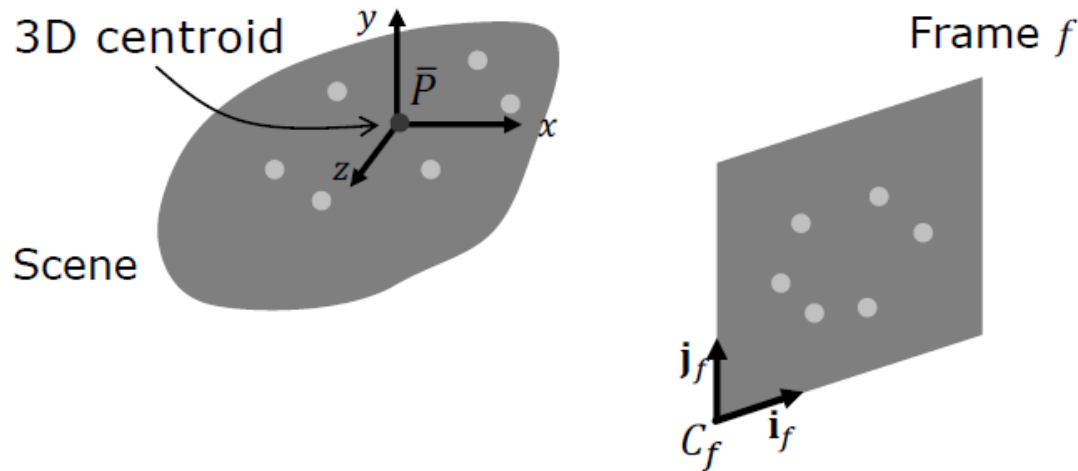
Image of point P_p in camera frame f :

$$\begin{aligned} u_{f,p} &= \mathbf{i}_f^T (P_p - C_f) \\ v_{f,p} &= \mathbf{j}_f^T (P_p - C_f) \end{aligned} \quad \boxed{1}$$

Known Unknown

We can remove C_f from equations to simplify SFM problem.

Centering Trick

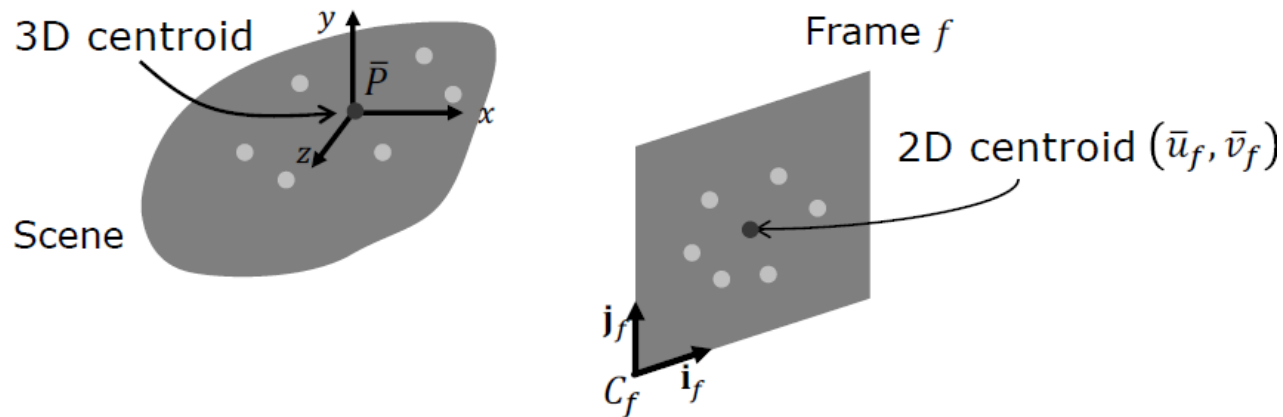


Assume origin of world at centroid of scene points:

$$\frac{1}{N} \sum_{p=1}^N P_p = \bar{P} = \mathbf{0} \quad \boxed{1}$$

We will compute scene points w.r.t their centroid!

Centering Trick



Centroid $(\bar{u}_f, \bar{v}_f)$ of the image points in frame f :

$$\bar{u}_f = \frac{1}{N} \sum_{p=1}^N u_{f,p} = \frac{1}{N} \sum_{p=1}^N \mathbf{i}_f^T (P_p - C_f) \quad [1] \quad \bar{v}_f = \frac{1}{N} \sum_{p=1}^N v_{f,p} = \frac{1}{N} \sum_{p=1}^N \mathbf{j}_f^T (P_p - C_f)$$

$$\bar{u}_f = \frac{1}{N} \cancel{\mathbf{i}_f^T \sum_{p=1}^N P_p} - \frac{1}{N} \sum_{p=1}^N \mathbf{i}_f^T C_f \quad [2] \quad \bar{v}_f = \frac{1}{N} \cancel{\mathbf{j}_f^T \sum_{p=1}^N P_p} - \frac{1}{N} \sum_{p=1}^N \mathbf{j}_f^T C_f$$

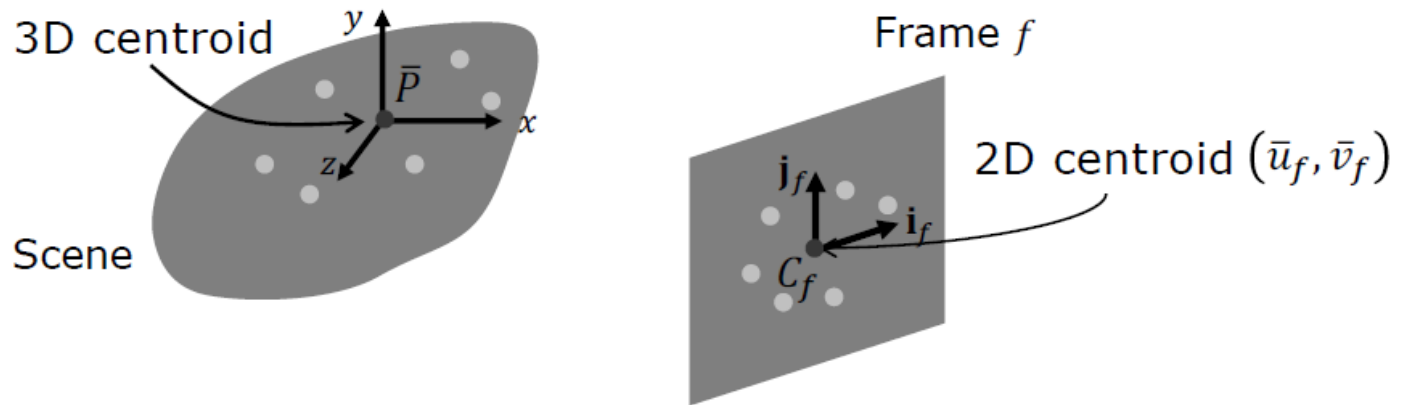
$$\boxed{\bar{u}_f = -\mathbf{i}_f^T C_f} \quad [3]$$

$$\boxed{\bar{v}_f = -\mathbf{j}_f^T C_f}$$

1

- The second term in 2 is an average computed over a constant.
- We see that the centroid in the frame is independent of the 3D scene points P_p .

Centering Trick



Shift camera origin to the centroid $(\bar{u}_f, \bar{v}_f)$.

Image points w.r.t. $(\bar{u}_f, \bar{v}_f)$:

$$\tilde{u}_{f,p} = u_{f,p} - \bar{u}_f$$

$$\tilde{v}_{f,p} = v_{f,p} - \bar{v}_f$$

$$= \mathbf{i}_f^T (P_p - C_f) + \mathbf{i}_f^T C_f$$

$$= \mathbf{j}_f^T (P_p - C_f) + \mathbf{j}_f^T C_f$$

$$\boxed{\tilde{u}_{f,p} = \mathbf{i}_f^T P_p} \quad \boxed{1}$$

$$\boxed{\tilde{v}_{f,p} = \mathbf{j}_f^T P_p} \quad \boxed{2}$$

Camera locations C_f now removed from equations.

1

- Now, image points only depend on the camera orientation, which is given by the two vectors $\mathbf{i}_f$ and $\mathbf{j}_f$.

Observation Matrix W

$$\boxed{1} \quad \begin{aligned} \tilde{u}_{f,p} &= \mathbf{i}_f^T P_p \\ \tilde{v}_{f,p} &= \mathbf{j}_f^T P_p \end{aligned} \quad \Rightarrow \quad \begin{bmatrix} \tilde{u}_{f,p} \\ \tilde{v}_{f,p} \end{bmatrix} = \begin{bmatrix} \mathbf{i}_f^T \\ \mathbf{j}_f^T \end{bmatrix} P_p \quad \boxed{2}$$

	Point 1	Point 2	...	Point N			Point 1	Point 2	...	Point N
Image 1	$\tilde{u}_{1,1}$	$\tilde{u}_{1,2}$	...	$\tilde{u}_{1,N}$	=	$\begin{bmatrix} \mathbf{i}_1^T \\ \mathbf{i}_2^T \\ \vdots \\ \mathbf{i}_F^T \\ \mathbf{j}_1^T \\ \mathbf{j}_2^T \\ \vdots \\ \mathbf{j}_F^T \end{bmatrix}$				
Image 2	$\tilde{u}_{2,1}$	$\tilde{u}_{2,2}$	...	$\tilde{u}_{2,N}$						
	$\vdots$	$\vdots$	$\vdots$	$\vdots$						
Image F	$\tilde{u}_{F,1}$	$\tilde{u}_{F,2}$	...	$\tilde{u}_{F,N}$						
Image 1	$\tilde{v}_{1,1}$	$\tilde{v}_{1,2}$	...	$\tilde{v}_{1,N}$						
Image 2	$\tilde{v}_{2,1}$	$\tilde{v}_{2,2}$	...	$\tilde{v}_{2,N}$						
	$\vdots$	$\vdots$	$\vdots$	$\vdots$						
Image F	$\tilde{v}_{F,1}$	$\tilde{v}_{F,2}$	...	$\tilde{v}_{F,N}$						

$W_{2F \times N}$
 Centroid-Subtracted
 Feature Points (Known)

$M_{2F \times 3}$
 Camera Motion
 (Unknown)

$S_{3 \times N}$
 Scene Structure
 (Unknown)

Rank of Observation Matrix

Will discuss the properties of W

Linear Independence of Vectors

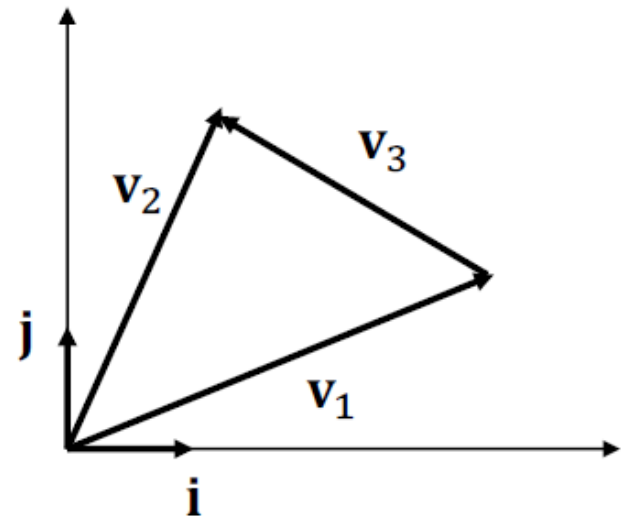
A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is said to be linearly independent if no vector can be represented as a weighted linear sum of the others.

$\{\mathbf{i}, \mathbf{j}\}$ is linearly independent.

$\{\mathbf{i}, \mathbf{j}, \mathbf{v}_1\}$ is linearly dependent.

$\{\mathbf{i}, \mathbf{j}, \mathbf{v}_3\}$ is linearly dependent.

$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent.



Rank of a Matrix

Column Rank: The number of linearly independent columns of the matrix.

Row Rank: The number of linearly independent rows of the matrix.

$$\begin{matrix} m \\ \left[\begin{array}{c} A \\ \end{array} \right] \\ n \end{matrix} = [\mathbf{c}_1 \quad \mathbf{c}_2 \quad \dots \quad \mathbf{c}_n] = \begin{bmatrix} \mathbf{r}_1^T \\ \mathbf{r}_2^T \\ \vdots \\ \mathbf{r}_m^T \end{bmatrix}$$

$$\text{ColumnRank}(A) \leq n$$

$$\text{RowRank}(A) \leq m$$

$$\text{ColumnRank}(A) = \text{RowRank}(A) = \text{Rank}(A)$$

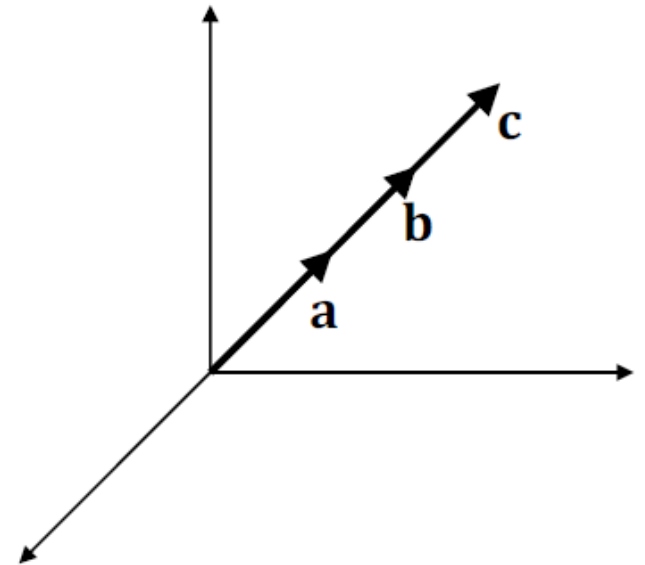
$$\text{Rank}(A) \leq \min(m, n)$$

Geometric Meaning of Matrix Rank

Rank is the dimensionality of the space spanned by column or row vectors of the matrix.

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} = [\mathbf{a} \quad \mathbf{b} \quad \mathbf{c}]$$

$$\text{Rank}(A) = 1$$

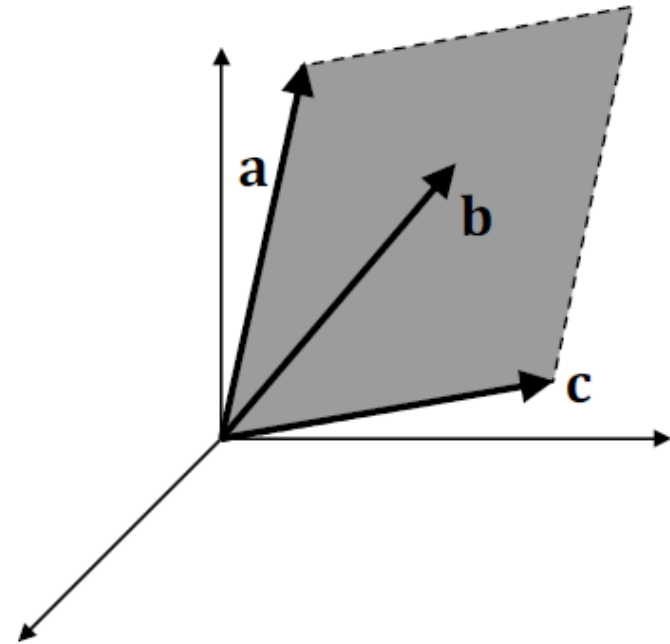


Geometric Meaning of Matrix Rank

Rank is the dimensionality of the space spanned by column or row vectors of the matrix.

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} = [\mathbf{a} \quad \mathbf{b} \quad \mathbf{c}]$$

$$\text{Rank}(A) = 2$$

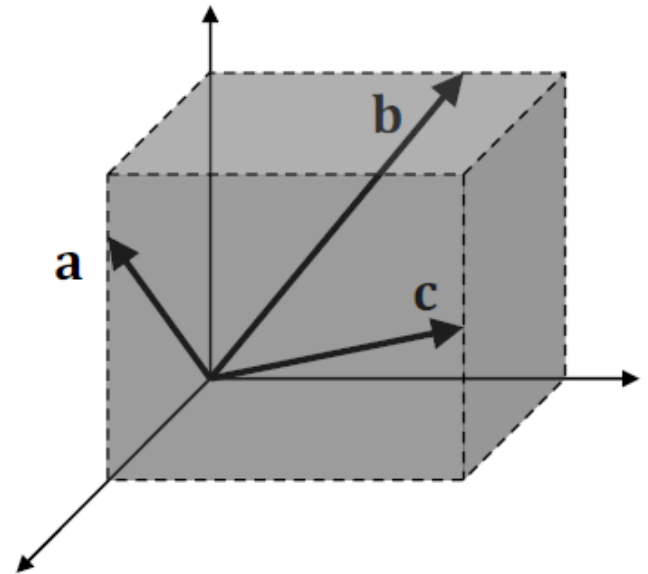


Geometric Meaning of Matrix Rank

Rank is the dimensionality of the space spanned by column or row vectors of the matrix.

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} = [\mathbf{a} \quad \mathbf{b} \quad \mathbf{c}]$$

$$\text{Rank}(A) = 3$$



Properties of Rank

- $\text{Rank}(A^T) = \text{Rank}(A)$
- $\text{Rank}(A_{m \times n} B_{n \times p}) \leq \min(\text{Rank}(A_{m \times n}), \text{Rank}(B_{n \times p}))$
 $\leq \min(m, n, p)$
- $\text{Rank}(AA^T) = \text{Rank}(A^T A) = \text{Rank}(A^T) = \text{Rank}(A)$
- $A_{m \times m}$ is invertible iff $\text{Rank}(A_{m \times m}) = m$ (Full Rank)

Tomasi-Kanade Factorization

Can we factorize W into M and S matrices

Back to Observation Matrix

$$\begin{array}{c}
 \text{Image 1} \\
 \text{Image 2} \\
 \vdots \\
 \text{Image F} \\
 \text{Image 1} \\
 \text{Image 2} \\
 \vdots \\
 \text{Image F}
 \end{array}
 \begin{array}{c}
 \text{Point 1} \\
 \text{Point 2} \\
 \vdots \\
 \text{Point N}
 \end{array}
 \begin{bmatrix}
 \tilde{u}_{1,1} & \tilde{u}_{1,2} & \dots & \tilde{u}_{1,N} \\
 \tilde{u}_{2,1} & \tilde{u}_{2,2} & \dots & \tilde{u}_{2,N} \\
 \vdots & \vdots & \vdots & \vdots \\
 \tilde{u}_{F,1} & \tilde{u}_{F,2} & \dots & \tilde{u}_{F,N} \\
 \tilde{v}_{1,1} & \tilde{v}_{1,2} & \dots & \tilde{v}_{1,N} \\
 \tilde{v}_{2,1} & \tilde{v}_{2,2} & \dots & \tilde{v}_{2,N} \\
 \vdots & \vdots & \vdots & \vdots \\
 \tilde{v}_{F,1} & \tilde{v}_{F,2} & \dots & \tilde{v}_{F,N}
 \end{bmatrix}
 =
 \begin{bmatrix}
 \mathbf{i}_1^T \\
 \mathbf{i}_2^T \\
 \vdots \\
 \mathbf{i}_F^T \\
 \mathbf{j}_1^T \\
 \mathbf{j}_2^T \\
 \vdots \\
 \mathbf{j}_F^T
 \end{bmatrix}
 \begin{array}{c}
 \text{Point 1} \quad \text{Point 2} \quad \dots \quad \text{Point N} \\
 [P_1 \quad P_2 \quad \dots \quad P_N] \\
 S_{3 \times N} \\
 \text{Scene Structure} \\
 \text{(Unknown)}
 \end{array}$$

$W_{2F \times N}$
 Centroid-Subtracted
 Feature Points (Known)

$M_{2F \times 3}$
 Camera Motion
 (Unknown)

Rank of Observation Matrix

$$\begin{array}{ccccc} W & = & M & \times & S \\ 2F \times N & & 2F \times 3 & & 3 \times N \end{array}$$

We know:

$$\text{Rank}(MS) \leq \text{Rank}(M) \qquad \text{Rank}(MS) \leq \text{Rank}(S)$$

$$\Rightarrow \text{Rank}(MS) \leq \min(3, 2F) \qquad \text{Rank}(MS) \leq \min(3, N)$$

$$\Rightarrow \text{Rank}(W) = \text{Rank}(MS) \leq \min(3, N, 2F)$$

Rank Theorem: $\text{Rank}(W) \leq 3$

We can “factorize” W into M and S !

Singular Value Decomposition (SVD)

For any matrix A there exists a factorization:

$$A_{M \times N} = U_{M \times M} \cdot \Sigma_{M \times N} \cdot V^T_{N \times N} \quad \boxed{1}$$

where U and V^T are orthonormal and Σ is diagonal.

MATLAB: `[U, S, V] = svd(A)`

$$\Sigma_{M \times N} = \begin{pmatrix} \sigma_1 & 0 & 0 & 0 & \dots & 0 \\ 0 & \sigma_2 & 0 & 0 & \dots & 0 \\ 0 & 0 & \sigma_3 & 0 & \dots & 0 \\ 0 & 0 & 0 & \sigma_4 & \dots & 0 \\ 0 & 0 & 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & 0 & \dots & \sigma_N \\ 0 & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots \end{pmatrix} \quad \sigma_1, \dots, \sigma_N: \text{Singular Values}$$

If $\text{Rank}(A) = r$ then A has r non-zero singular values.

SVD of W

Using SVD:

$$W = U \Sigma V^T$$

$$= \begin{pmatrix} & U & \end{pmatrix} \begin{pmatrix} \sigma_1 & 0 & 0 & 0 & \dots & 0 \\ 0 & \sigma_2 & 0 & 0 & \dots & 0 \\ 0 & 0 & \sigma_3 & 0 & \dots & 0 \\ 0 & 0 & 0 & \sigma_4 & \dots & 0 \\ 0 & 0 & 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & 0 & \dots & \sigma_N \\ 0 & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots \end{pmatrix} \begin{pmatrix} & V^T & \end{pmatrix}$$

$2F \times 2F$ $2F \times N$ $N \times N$

Where: $\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \dots \geq \sigma_N$ are the singular values of W

Enforcing Rank Constraint

Using SVD:

$$W = U \Sigma V^T$$

$$= \begin{pmatrix} U & \\ & \end{pmatrix}_{2F \times 2F} \begin{pmatrix} \sigma_1 & 0 & 0 & 0 & \dots & 0 \\ 0 & \sigma_2 & 0 & 0 & \dots & 0 \\ 0 & 0 & \sigma_3 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots \end{pmatrix}_{2F \times N} \begin{pmatrix} V^T & \\ & \end{pmatrix}_{N \times N}$$

Since $Rank(W) \leq 3$, $Rank(\Sigma) \leq 3$.

All except first 3 diagonal elements of Σ must be 0.

Enforcing Rank Constraint

Using SVD: $W = U \Sigma V^T$

$$= \begin{pmatrix} \begin{matrix} U_1 \\ U_2 \end{matrix} \end{pmatrix} \begin{pmatrix} \begin{matrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{matrix} & \begin{matrix} 0 & \dots & 0 \\ 0 & \dots & 0 \\ 0 & \dots & 0 \end{matrix} \\ \begin{matrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \vdots & \vdots & \vdots \end{matrix} & \begin{matrix} 0 & \dots & 0 \\ 0 & \ddots & 0 \\ 0 & \dots & 0 \\ 0 & \dots & 0 \\ \vdots & \dots & \vdots \end{matrix} \end{pmatrix} \begin{pmatrix} V_1^T \\ V_2^T \end{pmatrix}$$

$\begin{matrix} 3 & 2F-3 \\ 2F \times 2F \end{matrix}$
 $\begin{matrix} 2F \times N \end{matrix}$
 $\begin{matrix} N \times N \end{matrix}$

Since $\text{Rank}(W) \leq 3$, $\text{Rank}(\Sigma) \leq 3$.

Submatrices U_2 and V_2^T do not contribute to W .

Enforcing Rank Constraint

Using SVD:

$$W = U \Sigma V^T$$

$$= \begin{pmatrix} \begin{matrix} U_1 \\ U_2 \end{matrix} \begin{matrix} 3 & 2F-3 \end{matrix} \\ 2F \times 2F \end{pmatrix} \begin{pmatrix} \begin{matrix} \sigma_1 & 0 & 0 & 0 & \dots & 0 \\ 0 & \sigma_2 & 0 & 0 & \dots & 0 \\ 0 & 0 & \sigma_3 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots \end{matrix} \\ 2F \times N \end{pmatrix} \begin{pmatrix} \begin{matrix} V_1^T \\ V_2^T \end{matrix} \begin{matrix} 3 & N-3 \end{matrix} \\ N \times N \end{pmatrix}$$

$$\boxed{W = U_1 \Sigma_1 V_1^T} \quad \boxed{1}$$

(2F \times 3)(3 \times 3)(3 \times P)

Factorization: Finding M, S

$$W = \underbrace{U_1 (\Sigma_1)^{1/2}}_{(2F \times 3)} \underbrace{(\Sigma_1)^{1/2} V_1^T}_{(3 \times N)} \quad \boxed{1}$$
$$= M? \quad = S?$$

Not so fast. Decomposition not unique!

For any 3x3 non-singular matrix Q :

$$\boxed{2} \quad W = \underbrace{U_1 (\Sigma_1)^{1/2} Q}_{(2F \times 3)} \underbrace{Q^{-1} (\Sigma_1)^{1/2} V_1^T}_{(3 \times N)} \text{ is also valid.}$$
$$= M \quad = S \dots \text{ for some } Q.$$

How to find the matrix Q ?

Orthonormality of M

The Motion Matrix M :

$$M = \begin{bmatrix} \mathbf{i}_1^T \\ \vdots \\ \mathbf{i}_F^T \\ \mathbf{j}_1^T \\ \vdots \\ \mathbf{j}_F^T \end{bmatrix} = \underbrace{U_1(\Sigma_1)^{1/2}}_{\text{Computed}} Q = \underbrace{\begin{bmatrix} \hat{\mathbf{i}}_1^T \\ \vdots \\ \hat{\mathbf{i}}_F^T \\ \hat{\mathbf{j}}_1^T \\ \vdots \\ \hat{\mathbf{j}}_F^T \end{bmatrix}}_{\text{Computed}} Q = \begin{bmatrix} \hat{\mathbf{i}}_1^T Q \\ \vdots \\ \hat{\mathbf{i}}_F^T Q \\ \hat{\mathbf{j}}_1^T Q \\ \vdots \\ \hat{\mathbf{j}}_F^T Q \end{bmatrix} \quad \boxed{1}$$

Orthonormality Constraints:

$$\begin{aligned} \mathbf{i}_f \cdot \mathbf{i}_f &= \mathbf{i}_f^T \mathbf{i}_f = 1 \\ \mathbf{j}_f \cdot \mathbf{j}_f &= \mathbf{j}_f^T \mathbf{j}_f = 1 \\ \mathbf{i}_f \cdot \mathbf{j}_f &= \mathbf{i}_f^T \mathbf{j}_f = 0 \end{aligned}$$

➔

$$\begin{aligned} \hat{\mathbf{i}}_f^T Q Q^T \hat{\mathbf{i}}_f &= 1 \\ \hat{\mathbf{j}}_f^T Q Q^T \hat{\mathbf{j}}_f &= 1 \\ \hat{\mathbf{i}}_f^T Q Q^T \hat{\mathbf{j}}_f &= 0 \end{aligned}$$

2

Orthonormality of M

- We have computed $(\hat{\mathbf{i}}_f^T, \hat{\mathbf{j}}_f^T)$ for $f = 1, \dots, F$.

$$\left. \begin{aligned} \hat{\mathbf{i}}_f^T Q Q^T \hat{\mathbf{i}}_f &= 1 \\ \hat{\mathbf{j}}_f^T Q Q^T \hat{\mathbf{j}}_f &= 1 \\ \hat{\mathbf{i}}_f^T Q Q^T \hat{\mathbf{j}}_f &= 0 \end{aligned} \right\} Q \text{ is unknown.}$$

- Q is 3×3 matrix, 9 variables, $3F$ quadratic equations.
- Q can be solved with 3 or more images ($F \geq 3$) using Newton's method.

1

Final Solution:

$$M = U_1 (\Sigma_1)^{1/2} Q$$

Camera Motion

$$S = Q^{-1} (\Sigma_1)^{1/2} V_1^T$$

Scene Structure

Summary: Orthographic SFM

1. Detect and track feature points.
2. Create the centroid subtracted matrix W of corresponding feature points.
3. Compute SVD of W and enforce rank constraint.

$$W = U \Sigma V^T = U_1 \Sigma_1 V_1^T$$

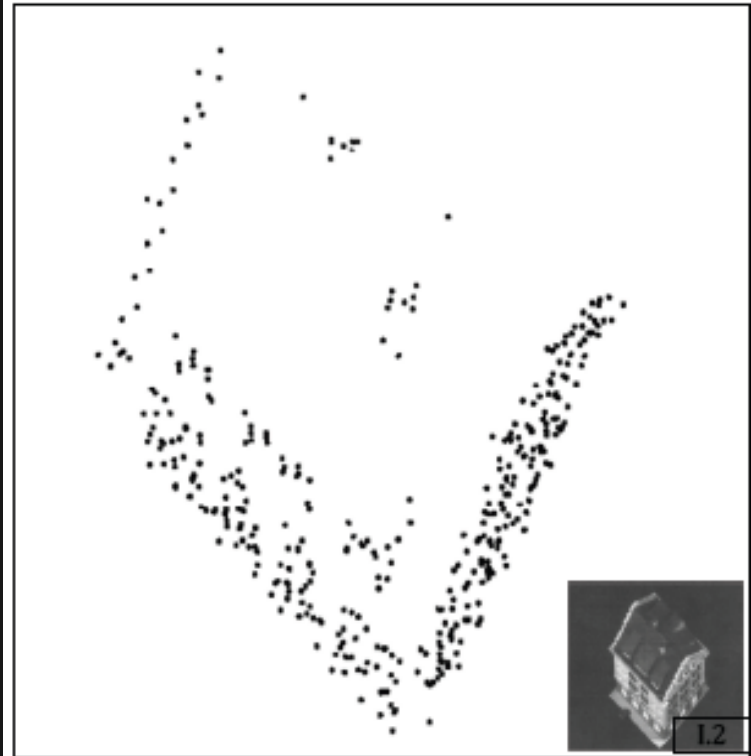
$(2F \times 3) \quad (3 \times 3) \quad (3 \times P)$

4. Set $M = U_1 (\Sigma_1)^{1/2} Q$ and $S = Q^{-1} (\Sigma_1)^{1/2} V_1^T$.
5. Find Q by enforcing the orthonormality constraint.

Results



Input Image Sequence

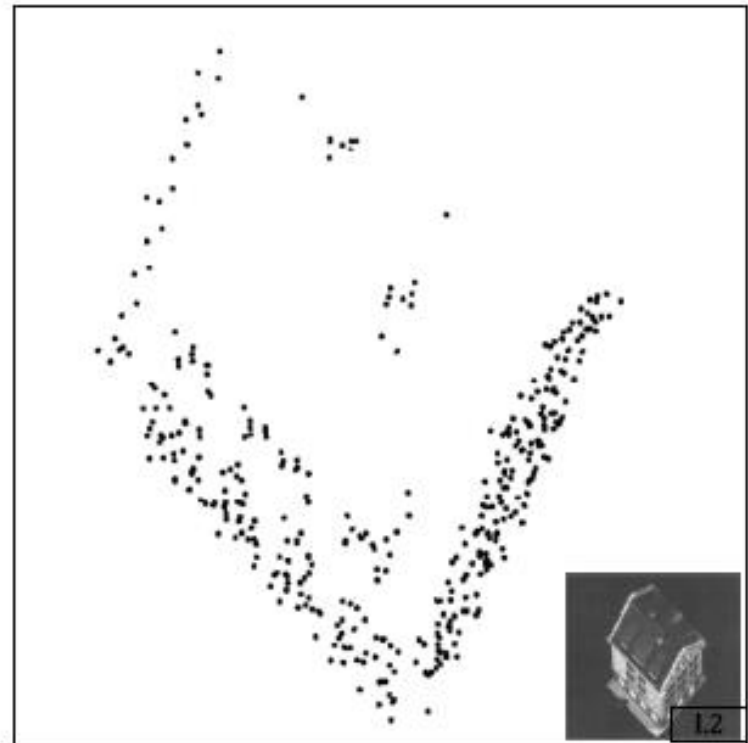


Estimated 3D Points

Results



Input Image Sequence



Estimated 3D Points

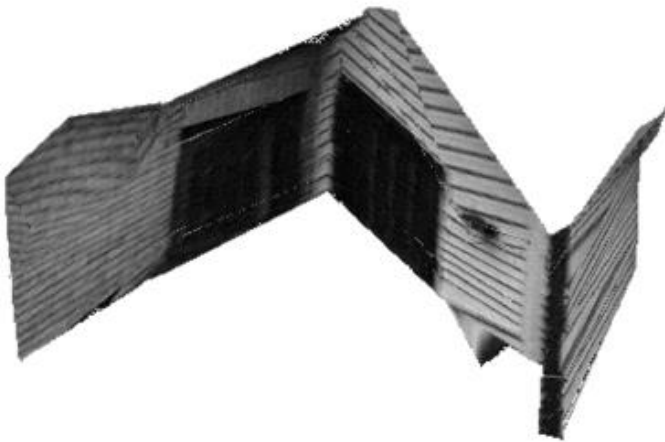
Results



Input Image Sequence



Tracked Features



3D Reconstruction



3D Reconstruction

SFM Results



- Pollefeys 2002

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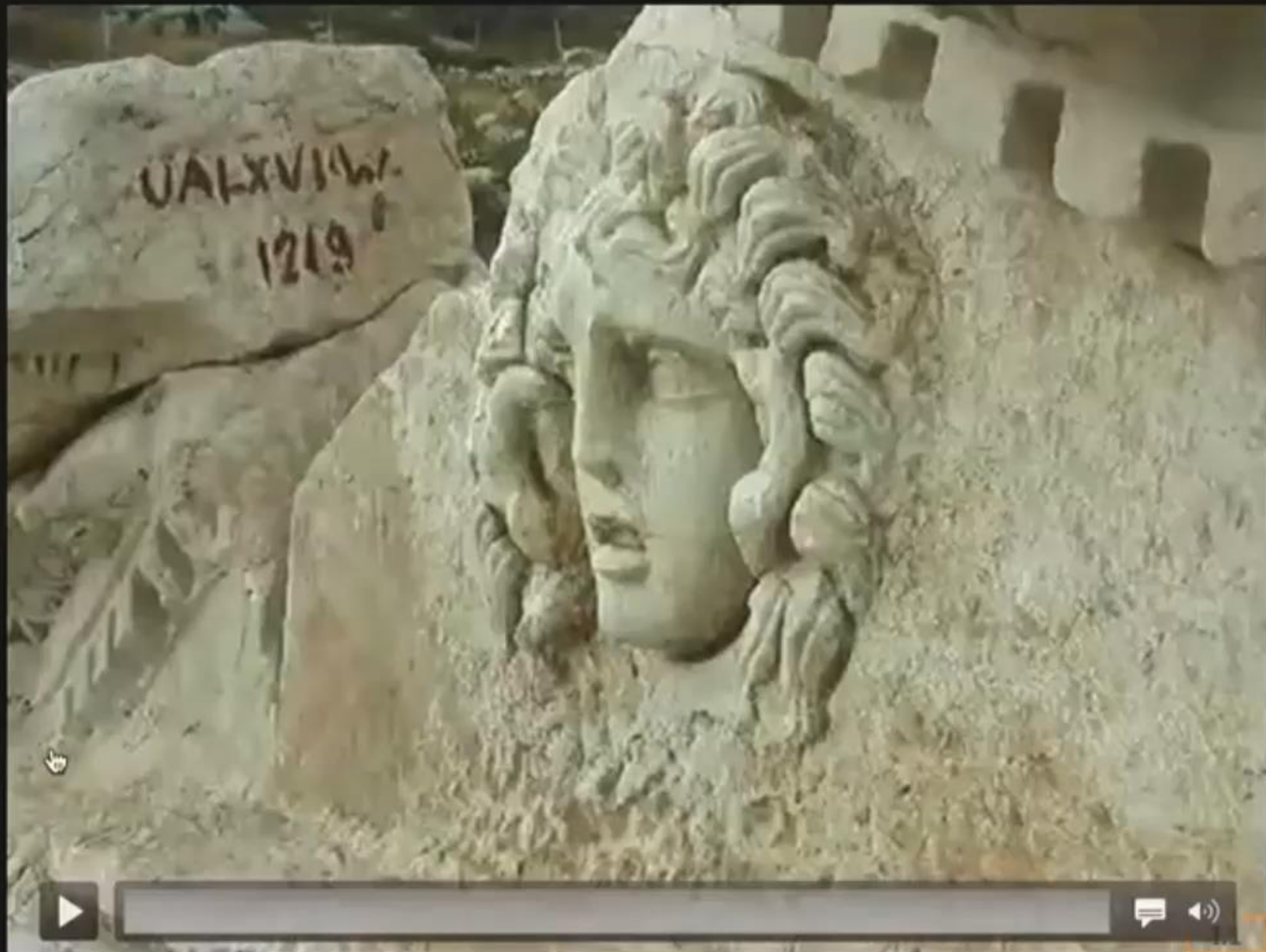
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[Pollefeys 2002] M. Pollefeys and L. Van Gool. Visual modeling: from images to images, The Journal of Visualization and Computer Animation, 13: 199-209, 2002.

SFM: Results



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 - J.B. Huang
 - Derek Hoiem
 - J. Hays
 - J. Johnson
 - R. Girshick
 - S. Lazebnik
 - K. Grauman
 - Antonio Torralba
 - Rob Fergus
 - Leibe
 - And many more